

IKRAMUL HASAN MORAL

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Final-year CSE student (CGPA 3.94) who builds production web platforms and does applied machine learning research. First-author paper under peer review, teaching assistant for 100+ students, two-time university champion.

EDUCATION

B.Sc. in Computer Science and Engineering | *United International University (UIU), Dhaka* Expected Dec 2026

- CGPA 3.94/4.0; 100% merit scholarship for 6 terms, awarded to the top 3% each trimester
- Thesis: Solving the Lorenz ODE System Using Optimal ANN Architectures; supervisor: Dr. Muhammad Nomani Kabir
- Coursework: Data Structures & Algorithms, Operating Systems, Computer Networks, DBMS, Software Engineering, Machine Learning, AI, HCI, Mobile Application Development, System Analysis & Design

TECHNICAL SKILLS

Languages: TypeScript, JavaScript, Python, C++, Java, SQL, HTML/CSS

Frontend: React 19, Next.js 15, React Native, Expo, Tailwind CSS, Radix UI, Zustand, TanStack Query, Redux

Backend: Node.js, Express, NestJS, MongoDB, PostgreSQL, Redis, Supabase, Firebase, REST, OpenAPI/Swagger, Zod, JWT, Socket.io, WebRTC

ML & Data: PyTorch, scikit-learn, XGBoost, SMOTE/imbalanced-learn, pandas, NumPy, SciPy, Matplotlib

Tools: Git, GitHub Actions, Docker, Vercel, Railway, Linux, Postman, Stripe, Cloudinary, Arduino

EXPERIENCE

Undergraduate Teaching Assistant | *United International University, Dept. of CSE* Aug 2025 - Present

- Teach Data Structures & Algorithms and DBMS to 100+ students; tutorials on graphs, DP, SQL, normalization
- Design and grade programming assignments; hold weekly office hours and code reviews

Freelance Full-Stack Developer | *EdTech client, IELTS e-learning platform (remote)* 2025 - Present

- Build and operate 3 live apps: Next.js 15 student portal, Express + TypeScript REST API, React 19 admin dashboard
- Integrated Vimeo/Zoom video streaming, AI study features, cloud file storage, and course/exam management
- Ship with Zod validation on every boundary, OpenAPI docs, JWT auth, server tests, and go-live checklists
- Long-term client relationship: rebuilt their first MERN site (2022) into today's production platform

RESEARCH

Handling Class Imbalance in UNSW-NB15 Intrusion Detection | *first author, under peer review* | [code](#) 2025 - 2026

- 18-experiment grid: binary + multiclass tasks; LR, RF, XGBoost; no balancing vs class weighting vs SMOTE
- Macro-F1, ROC-AUC, G-Mean with per-class analysis of rare attacks (Worms 0.07%, Shellcode 0.65%)
- Seeded, leakage-proof pipeline; preprocessing fit on train only; results reproducible from config snapshots

Undergraduate Thesis: Lorenz ODE System via Optimal ANN Architectures | *team lead of 5, FYDP-I defended* | [code](#) 2025 - Present

- 69-run controlled search: depth 1-4, width 20-100, 5 activations (tanh/ReLU/sigmoid/GELU/Swish), 3 optimizers
- Ground truth cross-verified with custom RK4 and SciPy DOP853 solvers; agreement RMSE $\sim 1.3e-11$
- 17-paper literature review; benchmarked against published PINN and DeepONet results

ANN Modeling of Hybrid Nanofluid Boundary Layer Flow | *independent scientific ML study* | [code](#) 2026

- 9-layer network trained with Levenberg-Marquardt optimization on $\sim 32,400$ physics-generated samples
- Validated against numerical solutions of the flow equations using MSE, RMSE, and R^2 metrics

PROJECTS

Advyon - Legal-Tech Platform | *React, Express, TypeScript, MongoDB* | [github](#) | [live](#) 2025

- Led 5-member team through full SDLC; Champion at UIU CSE Project Show (Software Engineering Lab)
- Case management, document workflows, community forum, smart intake, AI-assisted legal insights
- Express + TypeScript service layer with JWT auth, role-based access, and legal database services

BriefLi - AI Shortlink Manager | *React, TypeScript, Vite, Supabase, Vercel, Gemini API* | [github](#) | [live](#) 2025 - Present

- Live shortlink dashboard: AI slug suggestions, click tracking, public + authenticated link spaces, trash/restore
- Vercel serverless API; Supabase Postgres with SQL migrations and unique-slug constraints

Fall Detection Wearable | *Arduino/C++, Python ML, Node.js, JavaScript* | [github](#) 2025

- 4-layer IoT system: wearable firmware, Python ML fall classifier, Node.js alert server, caregiver web dashboard
- Real-time fall classification from accelerometer streams with instant alert delivery

Other Projects

- [VoiceKey](#) (offline Whisper voice typing, Linux/Windows) | [Paradox Prism](#) (anonymous reporting + steganography) | [PrintSlot](#) (campus print-shop app)
- [Skill Paradox](#) (skill-exchange platform) | [Meme Research Console](#) (live study instrument) | [Task Manager](#) (React Native/Expo) | [Portfolio](#) (Next.js 14, design-token theming, markdown blog) | [Email Templates](#) (live tool)

ACHIEVEMENTS & LEADERSHIP

- Champion & Team Lead, SE Lab, UIU CSE Project Show (Advyon); Champion, CSE Week Programming Contest 2024 (80+ participants)
- 500+ problems solved on [Codeforces](#), LeetCode, and HackerRank; active competitive programmer since 2022
- 1st Runner-Up, Advanced OOP Project Showcase 2023; 6th Runner-Up, System Analysis & Design Lab Showcase 2024
- Peer mentor, UIU CSE community; exam study packs used by classmates across the department; open-source contributor